

Solving Six-Player Games via Online Situation Estimation

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Abstract—While the artificial intelligence theory for solving the perfect-information games has been well developed in recent years, great challenges are still posed in dealing with the imperfect-information game due to the huge state space and hidden information involved in it. In this paper, we design an online strategy solving framework for six-player no-limit Texas hold'em poker. Based on hand isomorphism and hand strength evaluation, the framework provides an efficient situation estimation method for six-player poker. Such method could greatly reduce the state space in six-player poker as well as effectively evaluate the current hands. The poker agent based on our method won the third place in the 2018 AAAI-ACPC.

Index Terms—game, imperfect information, poker, six-player

I. INTRODUCTION

Computer games have always been regarded as the touchstone to verify the theory of artificial intelligence, and it is also one of the most attractive research domain of artificial intelligence. Generally speaking, the computer game can be divided into perfect-information game (PIG) and imperfect-information game (IIG). The PIG refers to the game in which each player could access to the accurate information, including characteristics, strategy spaces and profit functions, etc., of other players. The game of Go, as a typical PIG, presents such a situation that the positions of all chess pieces are visible to both of the players. By contrast, certain information is hidden in the IIG. Therefore each player holds specific information that cannot be overserved by other players. Since the IIG has such hidden information, it is much more difficult and challenging than the PIG [1].

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For a long time, Texas hold'em poker has been seen as the most representative game to research the IIG [1], [2], since its large state space and complex strategy. For example, heads-up no-limit hold'em (HUNL, one type of the Texas hold'em poker) is an extremely complex game with 10^{165} information sets. In the progress of playing, each player has to make decisions without knowing all the cards of other players, and meanwhile, tries to judge bluffing from opponents. The research about the IIG has attracted much attention in recent years. For example, counterfactual regret minimization (CFR) is a popular and effective method for solving the IIG [2]. Bowling proposes an algorithm framework for solving approximate Nash equilibrium using MC-CFR based on the state space compression [3]. DeepStack is designed for the IIG like poker [4], which beats professional human poker players in the HUNL in 44,000 hands. Brown et al. puts forward Libratus, which defeats four top human players in the 120,000 hands HUNL tournament. It solves the leading benchmark problem and the long-standing challenge of the IIG [5], [6].

Although the IIG agents like Deepstack and Libratus have shown high level intelligence in the field of the HUNL, such methods only designed for the two-player IIG [4], [6]. With the number of the player increases, the state space of the game will be explosively increasing, which makes the strategy solving of multi-player game become much more difficult. Despite some methods have been proposed to solve the strategy of the three-player poker [7], the multi-player poker is still a challenging problem in the field of the IIG.

This paper contributes to the study of decreasing the impact of large state space problem and increasing the efficiency

of obtaining the effective strategy. It does so by presenting a strategy solving framework for six-player no-limit Texas hold'em (SPNL), which is able to solve the online strategy in a limited time. A multi-player situation estimation method for six-player poker, which is based on the hand isomorphism and the hand strength evaluation, is introduced in SPNL. Such method could greatly reduce the state space in six-player poker as well as effectively evaluate the current hands. In addition, compared with methods that utilized with neural network, less history data of playing is required in our methods. The agent based on our method won the third place in the 2018 AAAI-ACPC (Annual Computer Poker Competition, a poker competition held by AAAI and IJCAI in turn) six-player no-limit Texas hold'em poker.

II. BACKGROUND

A. Zero-Sum Games

1) *Nash Equilibrium*: Nash equilibrium strategy or approximate Nash equilibrium strategy are usually used in the IIG with two players. It is a guideline about how to act on every information set that might arrive [8], [9]. The strategy σ_i of the player i for each information set $I_i \in L_i, \sigma_i(I_i) : A(I_i) \rightarrow [0, 1]$ is a probability distribution function in the information set $A(I_i)$. A strategy profile composed of the strategy of each player, which is the $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$. σ_{-i} means the strategy profile composed of the strategy of the remaining players except the player i .

Nash equilibrium is a strategy profile $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$ for n players. When the state of the game is in the Nash equilibrium, no participant can get a higher utility by unilaterally changing their strategy. The mathematical representation of the Nash equilibrium is as follows:

$$u_i(\sigma) \geq \max_{\sigma_i^* \in \Sigma_i} u_i(\sigma_i^*, \sigma_{-i}), \quad (1)$$

where u_i is the utility of the player i .

2) *Counterfactual Regret*: CFR(Counterfactual Regret Minimization) is a popular algorithm to solve the Nash equilibrium strategy, which is based on the minimization of regret values [2]. The core of the Minimization algorithm of regret value lies in the exploration of the Nash equilibrium [8].

The CFR contains three steps: Firstly, the regret minimization algorithm compares the utility of each strategy with the average utility and obtain the difference. Secondly, the algorithm selects the corresponding strategy next time according to the difference obtained before. Finally, the algorithm chooses the next appropriate strategy in the game according to the difference of size. It can be regarded as a balance if average values from both sides of players are less than the regret value. The method of calculating the average regret value is depicted as:

$$R_t^T = \frac{1}{T} \max_{\sigma_i^* \in \Sigma_i} \sum_{t=1}^T (u_i(\sigma_i^*, \sigma_{-i}^t) - u_i(\sigma^t)), \quad (2)$$

where T is the number of games played, and u_i is the counterfactual value.

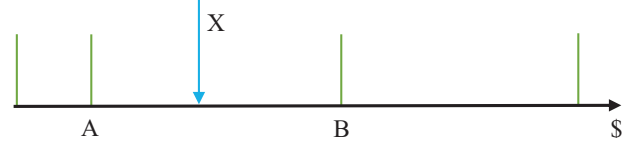


Fig. 1. Action mapping under Lipschitz's continuity

In recent years, CFR algorithm and its related variants are widely used to calculate the Nash equilibrium strategy in the IIG [2], [3], [5]. Although CFR and its variants have achieved good results in solving the Nash equilibrium, they are only solving two-player games.

B. Game Abstraction

Due to the huge state space of the IIG, the state space of the game need to be compressed in usual. Action abstraction is the common method to compress the state space for the no-limit poker.

1) *Action Abstraction*: In limit Texas hold'em, players can take three actions: fold, call, raise, and the number of chips for raising is fixed. However, in no-limit Texas hold'em poker, the number of raising chips is nearly equivalent to be continuous, which results in plenty of actions and a large breadth of game tree [9]–[11]. Clustering the betting actions of similar chips can greatly reduce the breadth of the game tree, and the betting chips according to different multiples of the current pot are clustered usually [12]–[16].

The strategy that is calculated after the action abstraction is applied to the original game. The action taken by the opponent has to be mapped to the abstract action set. The action mapping algorithm needs to meet the theoretical requirements of boundary constraint, monotonicity, scale invariance, robustness and boundary robustness. The algorithm based on the probability distribution used in Tartannian 5 is a common action abstraction algorithm [17].

Assuming that $S \in [T, T']$ is the legal chip in a given information set, raising chips of the opponents are x , and the abstract action set A, B satisfy $A = \max \{A_i : A_i \leq x\}$ and $B = \min \{A_i : A_i > x\}$. The problem of action mapping can be transformed into the problem of mapping x to A or B according to the probability. The probability distribution function satisfying Lipschitz's continuity is:

$$f_{A,B} = \frac{(B - x)(1 + A)}{(B - A)(1 + x)}, \quad (3)$$

as shown in Fig.1, the actual betting action is mapped to A or B according to the calculated probability.

III. OUR METHOD

The state space of the HUNL is about 10^{165} . The existing methods, such as DeepStack and Libratus, can deal with the HUNL [4], [6]. However, due to the huge state space of the SPNL which is much larger than that of the HUNL, the existing method is not suitable for this kind of games.

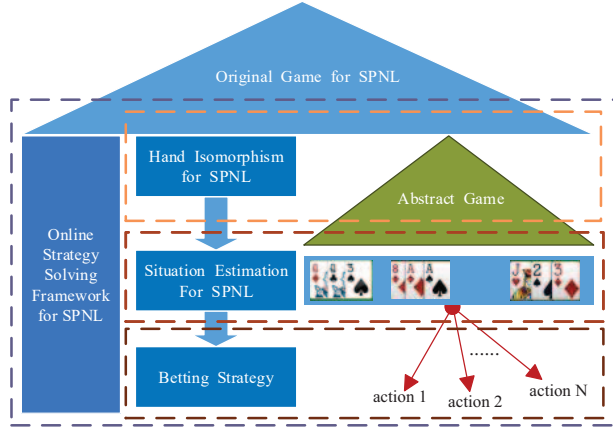


Fig. 2. The online strategy solving framework for SPNL

According to the characteristics of the six-player game, we propose a situation estimation method for six-player poker based on the hand isomorphism and the hand strength evaluation. There are three steps in our method. Firstly, the original game is abstracted into an abstract game with smaller state space, which maintains main properties of the original game. Secondly, the multi-player situation estimation method is used to evaluate the current state. Finally, the strategy of the abstract game is obtained by the betting strategy. The online strategy solving framework for SPNL is shown in Fig.2.

A. Hand Isomorphism for SPNL

In Texas hold'em, some hands will lead to the same strategy because there is no difference between different colors. In the pre-flop, for example, hA-s10, sA-d10 and dA-h10 should be considered as the same. That is, the corresponding solving strategy of these hands should be the same. Therefore, the size of the game tree can be greatly reduced by clustering the same type of hand combinations into the same kind. We refer to the hand isomorphism algorithm in [17], and design the algorithm specialized for the SPNL. Note that the values of the index of the hand with the same color are the same by indexing the hand of the player.

The algorithm that used to reduce the size of the state space is depicted in algorithm 1.

In the algorithm, the hand isomorphism algorithm indexes each hand. The index of the hand of pattern isomorphisms consists of four rounds: pre-flop, flop, turn and river. In each round, new cards are issued. Three steps are taken to index the hand:

Step 1. Index the hand in each round.

Step 2. Establish the value of the index of multi-round's hands based on the results of the index of the previous step.

Step 3. Introduce the concept of color configuration and establish the final value of the index.

B. Situation Estimation for SPNL

1) *Current hand strength*: The hand of the player is the combination of private cards and public cards. Thus the strength of the hand can be expressed as a probability. For example, Texas hold'em poker has 52 cards, and the hands of opponents can be represented by the remaining 50 cards in the pre-flop round. In this case, our hand is better(+1), tied(0) and worse(-1) compared with the hand strength of opponents. The hand strength can be computed through taking the summation and divided by the total number of possible opponents' hands. It is depicted as:

$$HS = \frac{N_{Better} - N_{Worse}}{N}, \quad (4)$$

where N is the number of possible poker situations. Compared with the opponents' hand strength, N_{Better} is the number of better cases and N_{Worse} is the number of worse cases.

Algorithm 1 The hand isomorphism algorithm for SPNL

Input: Hand $C_1, C_2, \dots, C_k$

Output: Index of hands

- 1: Map a hand to typical hand according to its pattern and color, $C'_1, C'_2, \dots, C'_k$
- 2: **for** i in $\{c, d, s, t\}$ **do**
- 3: Index the same color of the K wheel, $A_1; A_2, \dots, A_k$
- 4: $B \leftarrow A_1$
- 5: $next \leftarrow \text{indexgroup}_{m2, \dots, mk} (A_2, \dots, A_k \mid \cup A_1)$
- 6: $idx \leftarrow \binom{N-|u|}{|A_1|} next$
- 7: **for** i from 1 to $|A_1|$ **do**
- 8: $b \leftarrow \text{largest}(B)$
- 9: $rank \leftarrow b - |\text{smaller}(b) \cap V|$
- 10: $idx \leftarrow idx + \binom{rank}{|A_i|+i-1}$
- 11: Delete b from set B
- 12: **end for**
- 13: $Idx \leftarrow Idx + \binom{N-M+1}{M}$
- 14: **end for**
- 15: **return** Index

Supposing our hand is hA-rJ in the pre-flop, the opponent has 1225 kinds of cards (two out of 50 cards, assuming one opponent). In the flop, the public card is rQ-h6-d5, then the opponent's hand has 1081 kinds of cards. In this case, our hand strength is 0.585, that is to say, when our hand is hA-rJ, the chance of our hand better than opponent's hand is 58.5%.

2) *Current hand potential*: The strength of the combination of private cards and public cards are considered together in the most of times while calculating the current hand strength HS. Considering that the current hand strength does not distinguish the private cards from the public cards, and hence the player do not know whether the private cards, the public cards or both of them have an impact on the current hand strength HS. The current hand strength may be the combination of smaller private cards and larger public cards, or the combination of larger private cards and smaller public cards, while the HS is invariant. Moreover, there are two public cards after the flop round, which will also change the hand strength HS.

Therefore, it is necessary to calculate the hand potential of the current hand. The hand potential refers to the possibility

that the combination of the following public cards and private cards can improve the winning probability of the previous hand. The players have to identify the potential impact of public cards later. With public cards are issued, the positive potential will increase the probability of winning and vice versa.

3) *Effective hand strength*: Effective hand strength combines current hand strength and current hand potential, which gives an estimation of the winning probability of current hand strength compared with the opponents.

P_{Pot} is the number of times when the current hand strength is behind the opponents but ends up ahead the opponents. N_{Pot} is the number of times when the current hand strength is ahead the opponents but ends up behind the opponents.

$$Pr(win) = HS \times (1 - N_{Pot}) + (1 - HS) \times P_{Pot} \quad (5)$$

In many cases, when one has the best hands at the moment without considering the potential negative effects, the choice of betting is a good action. Hence, what players are interested in is that whether their hands are the best at the moment or the probability to become the best will be improved, which is measured by the hand strength. In the SPNL, the EHS is used to estimate the current hand strength, and the formula for the effective hand strength EHS is as follows:

$$EHS = HS + (1 - HS) \times P_{Pot}. \quad (6)$$

C. Betting Strategy for SPNL

In limit Texas hold'em poker, players can only take three actions: fold, call, and raise, the number of chips for raising is fixed. All the possible actions can be taken into account when solving Nash equilibrium since the branch of the game tree is small in the sequence of actions. In no-limit Texas hold'em, however, players always have a wide range of options. For example, compared with heads-up limit Texas hold'em, whose state space of is 10^{17} , the state space of HUNL is 10^{165} . The reason for the large space of no-limit Texas hold'em is that there are too many actions that the players can choose. The state space of SPNL is much larger than that of HUNL, since the number of the player in SPNL increasing.

There are many available actions for players in no-limit Texas hold'em, however, professional players usually take the action of raising multiples of the current pot. For example, when the current pot is 150, the number of chips raised by the player is 500. In this case, the action of raising 500 is considered as a huge betting action. If the current pot is 1500, the same action of raising 500 is considered to be a medium betting action. The same chips are regarded as different bets in different situations. To make an efficient decision in such a huge action space, a heuristic approach is proposed to make the betting action strategy. The number of chips in the paper is multiple of the current pot.

The basic betting strategy for the SPNL is the following four steps:

Step 1. Carry out the hand isomorphism and establish the value of the index.

Step 2. Calculate the EHS of the current hand relative to the hand of the other five opponents.

Step 3. Transform EHS into the probability of the action by using a set of betting rules and formulas : $Pr(fold)$, $Pr(call)$, $Pr(raise)$.

Step 4. Generate random numbers and use these numbers to select actions from the probability distribution.

IV. EXPERIMENTS

ACPC (Annual Computer Poker Competition) is a poker competition held by AAAI and IJCAI in turn. In this paper, we design a betting strategy for SPNL, and use ACPC as an experiment platform. SPNL is one of the ACPC-2018 competitions and the poker agent based on our method got the third place.

A. ACPC Communication

At the beginning of the game, the poker agent establishes a connection with the server through the specified port number and communicates with the server using TCP/IP protocol. In games, poker agents do not communicate directly, but are forwarded through the server side. The communication format between the poker agent and the server is specified as follows:

$\langle serverMessage \rangle := \langle matchState \rangle$

$\langle matchState \rangle := MATCHSTATE : \langle position \rangle \langle handNumber \rangle \langle betting \rangle \langle cards \rangle$

The string of communication is composed of 5 parts, each part is separated by colon. The meaning of each character is shown in Table I.

TABLE I
THE MEANING OF THE CHARACTER

Character	Meaning
Matchstate	Identifying this information is normol cammunication between the agent and the server.
Position	Representing the player's position, 0 indicates a big blind, and 1 indicates a small blind.
HandNumber	Indicates what inning is in the game
betting	Represents the action sequences of the players, c (call), f (fold), r (raise)
cards	Represents public cards and private cards. d, c, s and h indicate colors.

B. Experiment Setup

We conduct experiments on SPNL, which is a standard and new benchmark in 2018 ACPC. The players start each game with 20,000 chips and alternate positions after each game. Each player can choose to either fold, call, or raise on each of the four rounds of betting. Calling means the player places a number of chips in the pot that is equal to the opponents' share. Raising means the player adds more chips to the pot than the opponents' share. Players cannot raise beyond the 20,000 they start with. At the start of each game of SPNL, both players are dealt two private cards from a standard 52-card deck. Details of rules about SPNL in our experiment are depicted in Table II.

C. Experiment Result

Our experiment is to play with the poker agents in the ACPC SPNL. The agents come from the top six in 2018 ACPC SPNL competition. Because the participants of these related agents did not provide relevant open source code, nor did they publish relevant implementation details. Thus, the result we provide is the official comparison result of 2018 ACPC. Our agent runs with RAM of 256G and 64 CPU cores. In the experiments, the results are measured in the standard units used in this field: milli big blinds per hand (mbb/h).

TABLE II
EXPERIMENTS DETAILS

Hands Per Match	3000
Stack Sizes	200 big blinds (400 small blinds)
Bet Sizes	No limit
Blind Sizes	50/100
Blind Structure	Reverse blinds, no ascending blinds
Illegal Actions	Any illegal action is interpreted as a call
Time per hand	2 seconds

TABLE III
EXPERIMENT RESULTS IN 2018 ACPC

Agent	Results (mbb/h)
our agent	1298 +/- 189
PokerBot5	3336 +/- 265
Paco	1669 +/- 159
Slumbot	910 +/- 167
PhantomX	-2389 +/- 268
Feste	-4824 +/- 243

In the Table III, the result is composed of two values. The first value is the expected winnings, in thousandths of a big blind per hand, for the row player. The second value is the 95% confidence interval around the mean.

TABLE IV
OPPONENT AGENTS AND THEIR METHODS

Agent	Method
PokerBot5	online opponent modeling, Bayesian graphical modeling
Paco	MC self-play, neural networks, a hand-crafted policy
Slumbot	Targeted CFR [18]
PhantomX	pre-computed strategy generated through MCTS
Feste	CFR [2]

Table IV shows the the opponent agents and their methods in our experiments. To our knowledge, in these poker agents, PokerBot5 uses an online opponent modeling approach where the opponent strategies are modeled using Bayesian graphical modeling. Paco attempts to approximate a Nash equilibrium using a combination of Monte Carlo self-play and neural networks, as well as a hand-crafted policy. Slumbot uses Targeted CFR to approximate a Nash equilibrium. PhantomX uses a pre-computed strategy generated through Monte Carlo tree search. Feste attempts to approximate a Nash equilibrium using CFR.

V. CONCLUSION

This paper deals with solving the strategy of six-player no-limit Texas hold'em. We design a framework to solve the online strategy of six-player no-limit Texas hold'em. A situation estimation method is proposed to reduce the state space in the six-player poker and to efficiently evaluate the current hand strength. Our method offers a new idea to solve the SPNL, and meanwhile, compared with methods that utilized with neural network, less history data of playing is required in our methods.

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